

In-Depth Optical Analysis of the OM SYSTEM OM-1 Mark II with M.Zuiko 60mm f/2.8 Macro in the Nauticam NA-OM1 Housing

Technical System Profile and Component Integration

The integration of the OM SYSTEM OM-1 Mark II mirrorless camera with the M.Zuiko Digital ED 60mm f/2.8 Macro lens inside the Nauticam NA-OM1 underwater housing establishes a professional-grade imaging system optimized for underwater macro and super-macro photography.¹ Analyzing the optical performance of this system requires a comprehensive understanding of the physical properties of each component and how they interact when submerged.⁴

The OM SYSTEM OM-1 Mark II features a high-speed 20-megapixel stacked CMOS sensor, 1,053 cross-type phase-detection autofocus points, and a high-resolution 5.76-million-dot OLED electronic viewfinder.⁷ Crucial for macro work, its in-body image stabilization (IBIS) provides up to 7 EV of correction, or up to 8 EV when utilizing compatible stabilized lenses.⁷

The M.Zuiko Digital ED 60mm f/2.8 Macro lens represents a 120mm full-frame equivalent focal length in the Micro Four Thirds (MFT) format.⁸ The lens configuration consists of 13 elements in 10 groups, incorporating extra-low dispersion (ED), high refractive index (HR), and extra-high refractive index (E-HR) glass to eliminate chromatic aberrations.¹⁰ It utilizes a floating inner focus system with a Movie & Still Compatible (MSC) drive mechanism, ensuring that the physical length of the lens remains entirely static (82 mm) during focus adjustments.⁹

The Nauticam NA-OM1 housing accommodates this setup via the N85 port system.⁷ Machined from hard-anodized aluminum alloy, the housing is depth-rated to 100 meters and integrates an advanced vacuum check and leak detection monitoring system.³ It places critical camera dials within ergonomic reach of the handles, facilitating rapid adjustments to aperture, shutter speed, and focus underwater.²

System Component	Physical Parameter	Land Specifications (In-Air)	Submerged System Role
OM SYSTEM OM-1 Mark II	Camera Body	20MP Stacked CMOS, 1053 AF points ⁷	High-speed tracking, focus stacking engine ⁹

M.Zuiko 60mm f/2.8 Macro	Optical Lens	82 mm length, 185g, 46mm filter ⁹	120mm equivalent focal length prime ⁸
Nauticam NA-OM1 Housing	Pressure Vessel	307mm × 179mm × 118mm, N85 mount ⁷	100m depth rating, ergonomic control transfer ³
Nauticam Macro Port 65	Port Interface	79 mm length, M67 front thread ¹⁴	BK-7 glass boundary, dry-to-wet transition ¹⁴

Physics of Flat Port Refraction

Submerging a housing equipped with a flat port introduces a distinct optical boundary between two media with highly disparate refractive indices.⁵ While the air inside the housing has a refractive index of $n_{\text{air}} \approx 1.00$, seawater possesses a significantly higher refractive index of $n_{\text{water}} \approx 1.33$.¹⁶ When light rays pass from the water through the flat glass of the port and into the housing's internal air chamber, they undergo refraction.⁵

According to Snell's Law, light rays entering the flat boundary at any angle other than perpendicular (90° to the port surface) are bent away from the normal.⁵ This refraction induces three distinct physical and optical alterations in the imaging chain:

1. **Narrowing of the Field of View:** The diagonal angle of view of the 60mm lens is reduced by approximately 25% to 33%, changing from its nominal 20° in air to roughly 15° in water.⁴ This effectively narrows the field of view, simulating a longer focal length.⁴
2. **Virtual Image Creation:** Refraction projects a virtual image of the subject closer to the camera sensor than the subject's actual physical distance.¹⁸ The camera lens must focus on this closer, curved virtual image rather than the physical subject.¹⁸
3. **Pincushion Distortion and Aberrations:** Light rays passing through the outer perimeter of a flat port experience greater refraction angles than those passing through the center.¹⁸ This introduces mild pincushion distortion, progressive radial blur, and chromatic aberrations (color fringing) near the edges of the frame.⁵ Because macro lenses utilize a narrow field of view, these peripheral aberrations are largely mitigated compared to wide-angle systems, making flat ports the preferred choice for close-up imaging.⁶

Minimal Focusing and Working Distance Analysis

Calculating the exact limits of close-up focus underwater requires defining the physical spatial boundaries of the submerged imaging system.¹⁷ Two metrics must be calculated: the minimal focusing distance (MFD)—measured from the camera's sensor plane to the physical subject—and the minimal working distance (MWD)—measured from the front surface of the

housing's flat port glass to the physical subject.¹⁷

Derivation of Underwater Distances

On land, the M.Zuiko 60mm lens achieves focus at a minimum distance of 19.0 cm (190 mm) from the camera's sensor plane.⁹ Within the dry chamber of the Nauticam NA-OM1 housing, the physical distance from the sensor plane to the external face of the Nauticam Macro Port 65 glass is fixed at approximately 10.0 cm.¹⁷ On land, this leaves an available working distance from the port glass to the subject of approximately 9.0 cm¹⁷:

$$\text{MWD}_{\text{land}} = \text{MFD}_{\text{land}} - d_{\text{sensor-to-glass}} = 19.0 \text{ cm} - 10.0 \text{ cm} = 9.0 \text{ cm}$$

When the system is submerged, the light path is split into a dry segment (10.0 cm inside the housing) and a wet segment (the water column outside the port).¹⁷ Light traveling through the water column is refracted at the flat glass-to-air interface.⁵ The relationship between physical distance in water (d_{physical}) and apparent distance in air (d_{apparent}) is governed by the refractive index of water ($n_{\text{water}} \approx 1.33$)¹⁶:

$$d_{\text{apparent}} = \frac{d_{\text{physical}}}{n_{\text{water}}}$$

To achieve focus at the absolute mechanical limit of the lens's focusing helical, the virtual image formed by the flat port must align with the lens's native minimum land working distance (

$\text{MWD}_{\text{land}} = 9.0 \text{ cm}$).¹⁷ Therefore, the physical minimal working distance underwater ($\text{MWD}_{\text{underwater}}$) is calculated as follows¹⁷:

$$\text{MWD}_{\text{underwater}} = \text{MWD}_{\text{land}} \times n_{\text{water}} = 9.0 \text{ cm} \times 1.33 = 11.97 \text{ cm} \approx 12.0 \text{ cm}$$

This reveals that the **minimal physical working distance underwater is 12.0 cm (4.72 inches)** when measured from the front of the flat port glass to the subject.¹⁷

For comparison, if one measures working distance directly from the front of the lens barrel itself inside the housing, bypassing the housing's physical boundaries, the value is approximately **8.0 cm (3.15 inches)**.¹ This shorter distance reflects the space occupied by the internal air gap and the thickness of the optical BK-7 port glass.¹⁴

The underwater minimal focusing distance ($\text{MFD}_{\text{underwater}}$) is the sum of the dry internal distance and the physical wet distance¹⁷:

$$\text{MFD}_{\text{underwater}} = d_{\text{sensor-to-glass}} + \text{MWD}_{\text{underwater}} = 10.0 \text{ cm} + 12.0 \text{ cm} = 22.0 \text{ cm}$$

Consequently, the **minimal focusing distance underwater is 22.0 cm (8.66 inches)** from the camera's sensor plane to the physical subject.¹⁷

Physical Implications for the Photographer

Understanding these shifted physical boundaries is critical for practical field operations.¹ A physical working distance of 12.0 cm from the port glass provides a significant advantage when photographing skittish or highly mobile marine subjects, such as gobies, blennies, or cleaner shrimp.¹ This distance reduces the likelihood of startling the subject or damaging delicate reef structures.¹

Furthermore, this physical gap provides sufficient clearance around the port to position external strobes or video lights, ensuring that the housing itself does not cast a shadow over the subject.¹

Magnification and Frame-Filling Dynamics

The M.Zuiko 60mm f/2.8 Macro lens is designed to deliver a native maximum reproduction ratio of 1:1 (1.0x magnification) on land, projecting a subject onto the sensor at its physical size.⁸

Effective Underwater Magnification

Submerged flat port refraction introduces a physical magnification factor of $1.33\times$.¹⁹ Since the camera focuses on a virtual image that is magnified by the water-air interface, the effective magnification of the physical subject on the sensor increases proportionally²¹:

$$\text{Magnification}_{\text{underwater}} = \text{Magnification}_{\text{native}} \times n_{\text{water}} = 1.0 \times 1.33 \approx 1.33\times$$

This corresponds to an underwater reproduction ratio of **1.33:1**.²¹ Due to the 2.0x crop factor of the Micro Four Thirds sensor format relative to 35mm full-frame sensors, this system achieves a **35mm equivalent magnification of 2.66x**.⁸

An ongoing debate in underwater optical physics concerns the nature of this magnification.²¹ Because the lens's focusing helical reaches its physical limit at 1:1, the physical image projected onto the sensor cannot exceed the physical dimensions of the sensor.²¹ However, because flat port refraction compresses the field of view, a smaller physical area of the subject fills the frame, yielding a practical, frame-filling magnification of 1.33x.¹⁸

Sensor Coverage and Subject Dimensions

On land, the minimum field size required to fill a Micro Four Thirds sensor (

$13.0 \text{ mm} \times 17.3 \text{ mm}$) at 1:1 magnification is $13.0 \text{ mm} \times 17.0 \text{ mm}$.¹⁰ Due to the 1.33x refractive magnification underwater, the minimum field size required to fill the frame decreases ¹⁰:

$$\text{Field Width}_{\text{underwater}} = \frac{17.0 \text{ mm}}{1.33} \approx 12.8 \text{ mm}$$

$$\text{Field Height}_{\text{underwater}} = \frac{13.0 \text{ mm}}{1.33} \approx 9.8 \text{ mm}$$

This means an underwater subject measuring just $9.8 \text{ mm} \times 12.8 \text{ mm}$ will completely fill the frame on the OM-1 Mark II sensor.¹⁰ This enables high-detail captures of minute marine organisms without the immediate necessity of external wet diopters.¹

Nauticam Housing and Port Architecture

To achieve these physical and optical baselines, the system must be configured using compatible housing components designed to match the physical dimensions of the M.Zuiko 60mm lens.² The Nauticam NA-OM1 housing relies on the compact N85 Port System.⁷ Nauticam provides specific configurations to support the M.Zuiko 60mm f/2.8 Macro lens, ensuring the front of the lens is aligned with the port glass to prevent vignetting and optimize the optical path.⁴

Port Configuration Option	Part Number	Mechanical Specifications and Optical Path Details
Dedicated Flat Port	36163 ¹⁵	N85 Macro Port 65: Direct-mount flat port. Features BK-7 optical glass (90mm diameter, 8mm thickness) and hard-anodized aluminum body with an M67 threaded front. ¹⁴ Length: 79 mm. ¹⁴
Modular Port Assembly	36135 & 36620 ²	N85 Macro Port 45 combined with N85 Mini Extension Ring 20: Achieves a matching 79

		mm length, replicating the dedicated Port 65 optical path. ²
Manual Focus Gear	36149 ³¹	Nauticam O60-F Focus Gear: Clamps directly to the lens's physical focus ring, enabling manual focus control via the housing's left-side dial. ¹⁴

Autofocus Engineering and Housing Ergonomics

The OM-1 Mark II's 1,053 cross-type phase-detection points are highly responsive.⁷ However, suspended particles in the water (backscatter) and low-contrast underwater environments can trigger focus hunting.²⁵

Focus Limiter Utilization

The M.Zuiko 60mm lens features a physical focus limit switch with four zones: 0.19m–infinity, 0.4m–infinity, 0.19m–0.4m, and 1:1.⁸ Submerged refraction shifts these limits. The 0.19m–0.4m setting physically translates to a subject-to-sensor range of approximately 22 cm to 53 cm in water. Setting the lens to the 0.19m–0.4m limit is highly recommended, as it prevents the AF system from searching into the water column beyond 53 cm, significantly accelerating focus acquisition.⁸

Back-Button Focus Integration

Utilizing a thumb-operated lever on the NA-OM1 housing to actuate the AF-ON control separates focusing from shutter release.² This technique allows the photographer to lock focus and then make minute physical body adjustments (rocking forward or backward) to establish critical focus on the subject's eye.²⁷ This is particularly useful when shooting with shallow depth of field.³³

Wet Close-Up Lenses and Focus Optimizers

For subjects smaller than ¹⁰ mm, the M67 front thread on the Macro Port 65 allows the attachment of wet close-up lenses (diopters) underwater.⁴ These lenses shorten the system's focal length, allowing the camera to focus closer and increase magnification.⁴

Wet Optic Accessory	Optical Purpose	Magnification with 60mm Lens	Practical Working Distance
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Nauticam CMC-2 ³⁵	Moderate magnification, easier handling	~1.7x ³⁶	3.3 cm to 12.2 cm ³⁶
Nauticam CMC-1 ³⁷	High magnification for tiny details	~2.3x ²⁸	2.0 cm to 5.0 cm ¹⁷
Nauticam SMC-3 ³⁸	Ultra-low dispersion glass super-macro	~2.3x ³⁸	5.0 cm to 10.0 cm ³⁸
Nauticam MFO-1 ²⁵	Aberration correction and AF optimization	~1.1x ³⁶	Up to 1.14 meters ²⁵
Nauticam MFO-3 ³⁹	Mid-range focus optimizer for longer lenses	~1.1x ³⁹	Optimized for multi-flip configurations ⁴⁰

Mid-Range Focus Optimizers

The Nauticam Mid-range Focus Optimizer (MFO-1) is highly effective for fast-moving subjects.²⁵ Designed at the request of marine photographers, the MFO-1 corrects the optical aberrations (chromatic aberration and pincushion distortion) caused by flat ports.²⁵ By optimizing the light path, it increases the maximum working distance of the lens, eliminates autofocus hunting, and improves corner-to-corner sharpness.²⁵

Conclusions and Technical Recommendations

To maximize the optical performance of this system underwater, several technical practices are recommended:

- **Focus Limiter Constraints:** Set the lens's physical focus limit switch to the 0.19m–0.4m setting before sealing the housing.⁸ This constrains the autofocus mechanism to its optimal close-up range, preventing focus hunting in open water.⁸
- **Back-Button Autofocus Configuration:** Program the camera's AF-ON control to a housing thumb lever.² This allows the photographer to lock focus and utilize the "lock and rock" technique, moving the housing slightly forward and backward to achieve precise focus on the subject's eye.²⁷
- **Aperture Management:** To minimize the effects of diffraction while maintaining sufficient depth of field, operate within an aperture range of $f/5.6$ to $f/11$.⁹ For extremely small subjects where a larger depth of field is required, leverage the OM-1 Mark II's in-camera focus stacking capability.⁹

- **Wet Optic Selection:** For subjects under 10 mm, attach a Nauticam CMC-2 to achieve a 1.7x magnification ratio.³⁶ For larger, skittish subjects, utilize the Nauticam MFO-1 to minimize flat port aberrations and expand the effective working distance.²⁵

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